

School of Mathematics and Statistics

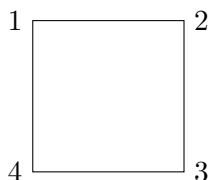
MT5864 Advanced Group Theory

Problem Sheet VII: Primitive Permutation Groups

1. Let G be a group that acts upon the set Ω , let Δ be a block for G and let $\delta \in \Delta$. Show that Δ is a union of orbits of the stabilizer G_δ .
2. Let G be a group that acts transitively on a set Ω . A G -congruence on Ω is an equivalence relation $\sim$ such that for all $\alpha, \beta \in \Omega$ and $x \in G$

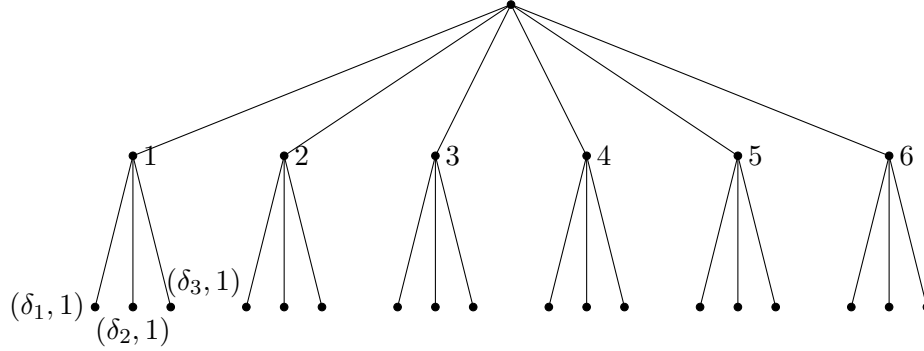
$$\alpha \sim \beta \quad \text{if and only if} \quad \alpha^x \sim \beta^x.$$

- (a) Suppose that Δ is a block for G and $\Sigma = \{\Delta^x \mid x \in G\}$ is the associated system of blocks. Define $\alpha \sim \beta$ if and only if α and β belong to the same set Δ^x . Show that $\sim$ is a G -congruence on Ω .
 - (b) Conversely if $\sim$ is a G -congruence on Ω , show that the equivalence classes of $\sim$ are blocks for G .
3. (a) Consider the action of the dihedral group D_8 of order 8 on the set $\{1, 2, 3, 4\}$ determined by the group's action on the labelled square:



- Determine all blocks for D_8 that contain 1.
- (b) Consider the natural action of the dihedral group D_{12} of order 12 on the set $\{1, 2, 3, 4, 5, 6\}$ obtained from the corresponding clockwise labelling of the vertices of a regular hexagon. Determine all blocks for D_{12} that contain 1.
4. Let G be a group and H be a subgroup of G . Consider the permutation representation $\rho: G \rightarrow \text{Sym}(\Omega)$ determined by the action of G on the set Ω of cosets of H .
 - (a) Under what condition(s) is this action faithful?
 - (b) If G is a simple group and $H \neq G$, is this action faithful?
 - (c) Under what condition(s) is this action primitive?

5. Let $\Sigma = \{1, 2, \dots, n\}$ and Δ be any finite set (where we assume $n \geq 2$ to avoid trivial cases). Consider the following tree $\mathcal{T}$:



This tree has a root (the single vertex at the top of the diagram), one vertex attached to the root for each element in Σ (and we have labelled each vertex by the corresponding element of Σ) and, to each vertex in the middle row, one vertex in the bottom row for each element in Δ (we label the vertices in the bottom row by pairs (δ, i) for $\delta \in \Delta$ and $i \in \Sigma$). [For space reasons, in the picture only the vertices in the bottom row that are attached to the vertex labelled 1 are labelled. In the diagram, $|\Sigma| = 6$ and $|\Delta| = 3$, but you should assume n and Δ are arbitrary in this question.]

Let $W = \text{Sym}(\Delta) \text{ wr }_{\Sigma} \text{Sym}(\Sigma)$ be the wreath product of the symmetric group on Δ by the symmetric group on Σ and consider its imprimitive action on $\Omega = \Delta \times \Sigma$:

$$(\delta, i)^{(h_1, h_2, \dots, h_n)k} = (\delta^{h_i}, ik).$$

- Show that this formula defines an automorphism of the above tree $\mathcal{T}$.
- Show that any automorphism of $\mathcal{T}$ is determined by the application of an element of the wreath product W .
- Deduce that $\text{Aut } \mathcal{T} \cong W$.

[Definition: An automorphism of a tree T is a bijection from the sets of vertices and of edges of the tree to themselves such that two vertices are joined in T if and only if their images are joined in T ; that is, it is bijection from the tree to itself that preserves the structure of the graph.]

6. Suppose that $\phi: \Omega \rightarrow \Omega'$ is a bijection. Show that the map $\theta: \text{Sym}(\Omega) \rightarrow \text{Sym}(\Omega')$ given by

$$\theta: \sigma \mapsto \phi^{-1} \sigma \phi$$

is an isomorphism between these symmetric groups.

7. Let G be a group and N be a normal subgroup. Let G act on N by conjugation and let $\rho: G \rightarrow \text{Aut } N$ be the corresponding permutation representation. Show that $\ker \rho = C_G(N)$, the centralizer of N in G .

(It follows that $C_G(N) \trianglelefteq G$ and that the quotient $G/C_G(N)$ is isomorphic to a subgroup of $\text{Aut } N$.)

8. Let G be an almost simple group with minimal normal subgroup N . Show that

- (a) the group $\text{Inn } N$ of inner automorphisms of N is isomorphic to N ;
- (b) there is a subgroup $\tilde{G}$ of the automorphism group of N with

$$\text{Inn } N \leq \tilde{G} \leq \text{Aut } N$$

such that $\tilde{G} \cong G$.

9. Let V be a vector space of dimension $n \geq 1$ over the finite field $\mathbb{F}_p$ (for some prime p) and let H be a subgroup of the general linear group $\text{GL}_n(\mathbb{F}_p)$. Let $G = V \rtimes H$ viewed as a subgroup of the affine general linear group $\text{AGL}_n(\mathbb{F}_p)$.

- (a) Show that the induced action of G on V is transitive.
- (b) Show that the stabilizer of 0 in G is $G_0 = \{ (0, h) \mid h \in H \}$.
- (c) Show that the action of G on V is primitive if and only if H acts irreducibly on V .
[Hint: If $G_0 < K \leq G$, consider $W = \{ v \in V \mid (v, 1) \in K \}$.]

[Definition: We say H acts *irreducibly* on the vector space V if there is no non-zero proper subspace W of V such that $wh \in W$ for all $w \in W$ and $h \in H$.]

10. Let $H \leq \text{Sym}(\Delta)$ and $K \leq \text{Sym}(\Sigma)$ where $\Sigma = \{1, 2, \dots, n\}$, $|\Delta| > 1$ and $n \geq 2$. Let

$$\Omega = \Delta^{(n)} = \{ (\delta_1, \delta_2, \dots, \delta_n) \mid \delta_1, \delta_2, \dots, \delta_n \in \Delta \}.$$

Define an action of the wreath product $W = H \text{ wr}_\Sigma K$ on Ω by

$$(\delta_1, \delta_2, \dots, \delta_n)^g = (\delta_{1k^{-1}}^{h_{1k^{-1}}}, \delta_{2k^{-1}}^{h_{2k^{-1}}}, \dots, \delta_{nk^{-1}}^{h_{nk^{-1}}})$$

for $g = (h_1, h_2, \dots, h_n)k$.

(So $(\delta_1, \dots, \delta_n)^{(h_1, \dots, h_n)} = (\delta_1^{h_1}, \dots, \delta_n^{h_n})$ specifies the action of the base group B and then elements of K permute the entries of the elements in Ω .)

- (a) Verify that this is an action of W on Ω .
(It is called the *product action* of the wreath product and, under sufficient assumptions, provides the “product type” example of a primitive permutation group appearing in the O’Nan–Scott Theorem.)
- (b) Show that the product action of W on Ω is faithful.
- (c) If H acts transitively on Δ , show that the product action of W on Ω is transitive.

For the remainder of the question, assume H acts transitively on Δ .

- (d) If $\delta \in \Delta$, find the stabilizer W_δ in W of the point $\delta = (\delta, \delta, \dots, \delta) \in \Omega$.
- (e) If $\delta \in \Delta$ and $H_\delta < L < H$ for some subgroup L , show that $R = L \text{ wr}_\Sigma K$ is a subgroup of W satisfying $W_\delta < R < W$.
- (f) Define $D = \{ (h, h, \dots, h) \mid h \in H \}$. If H acts regularly on Δ show that $DK = \{ dk \mid d \in D, k \in K \}$ is a subgroup of W satisfying $W_\delta < DK < W$ for any $\delta \in \Delta$.
- (g) Suppose J is an orbit of K on Σ with $J \neq \Sigma$. Define, for $\delta \in \Delta$,

$$M = \{ (h_1, h_2, \dots, h_n) \mid h_j \in H_\delta \text{ for all } j \in J \}.$$

Show that MK is a subgroup of W satisfying $W_\delta < MK < W$.

[Parts (e)–(g) show that if H acts imprimitively or regularly on Δ or if K acts intransitively on Σ , then the stabilizer of some point in Ω is *not* maximal. Hence if the product action of W on Ω is primitive, then H acts primitively but not regularly on Δ and K acts transitively. In fact, these conditions characterize (for finite permutation groups) when the product action of the wreath product is primitive.]